

Perfected plan authority and delta

Authoritative plan: docs/winning-plan/FireAtlas_Updated_Perfected_Winning_Plan.pdf

Superseded planning snapshot: docs/winning-plan/FireAtlas_Winning_Plan.pdf

Received: 30 September 2026

This source file makes the supplied perfected plan the working specification. The earlier PDF remains useful as a record of decisions already made, but its score ceiling and task order must not be treated as a promise. A score changes only when the evidence gate below is passed.

What the perfected plan changes

New requirement	What the repository can claim today	Gate still open
Primary unit is VIIRS-equivalent active-fire cell-days on a common 1 km grid ; native VIIRS 375 m remains detail.	The calendar and Sensor Bridge label the common-grid unit; imported rows retain native source fields. The current transform is detection-centroid binning, not a native fire-mask resampling.	Native-mask/common-grid agreement and an independent review.
Show MODIS-only, VIIRS-only, both, and gap/unknown states.	The Sensor Bridge and calendar states expose these categories for the imported source-export window.	Complete archive windows and pass/cloud evidence for stronger coverage claims.
Show raw Fire Radiative Power separately in MW/day.	Raw FRP is shown per source and is never added to cell-days or combined across sensors.	Check every displayed total against the current export after any data refresh.
Add MCD64A1 Collection 6.1 as lagged burned-area corroboration.	The UI and method state ACTIVE FIRE ONLY and link the NASA product. No MCD64A1 value or overlay is fabricated.	Acquire, document, and independently review MCD64A1 data before showing corroboration.
Distinguish observed, scaled, unknown, documented processing gap, and complete zero export.	Existing calendar and evidence views preserve documented gaps, scaled values, unknown partial exports, and zero-export limitations. A scaled outage day carries <code>evidence_state=scaled</code> plus <code>coverage_state=documented_processing_gap</code> , and is written as “gap · scaled estimate”.	Verify the state on every complete/partial window and keep pass/cloud opportunity unknown without masks.
Provide a shareable evidence card.	The local calendar can create a card and a parameterized link from the same API/static JSON as the selected result.	Verify the link and card on the public host.
Keep Rapid Pulse separate from historical science.	The historical calendar does not silently mix near-real-time records; a separate rapid product is deferred.	Build only after P0/P1 and give it a separate mode, calibration, and limits.
Demonstrate See → Compare → Verify → Share, with a 30-second cut and a 90-second narrated cut.	The local storyboard and browser checks cover the existing globe, calendar, evidence drawer, method audit, and local card.	Capture the new bridge/card sequence and rehearse it on the public URL.

Non-negotiable scientific rules

1. Never compare raw MODIS 1 km pixels directly with raw VIIRS 375 m pixels.
2. Deduplicate within each sensor and UTC day before aggregation.
3. Never add both sensors into one activity count. Use one selected reference observation and show the other sensor for comparison.
4. Keep FRP as a separate source-context measure. It is not burned area or fire severity.
5. A complete zero export means zero eligible rows in that export. It does not prove a clear pass, cloud-free observation, or no fire.
6. Missing, incomplete, or documented-outage inputs remain unknown or gap-marked.
7. MCD64A1 is lagged burned-area context, never active-fire ground truth for the same day.
8. Historical claims must show product version, request coverage, retrieval/hash provenance, and the uncertainty interval. A weak held-out interval stays labelled weak.
9. The existing globe, its controls, its overlays, and its handoff remain unchanged.

Release gates under the perfected plan

P0 — required before a broader historical claim

- Recover missing standard archive windows for both study regions.
- Preserve request metadata, product version, UTC range, bounding box, retrieval date, SHA-256, row count, and excluded type counts.
- Reproduce common-grid daily and monthly totals from the source rows after deduplication.
- Keep all partial and undocumented windows unknown; do not fill gaps with generated values.

P1 — highest-value judge path

- Sensor Bridge with MODIS-only / VIIRS-only / both / gap states.
- Separate raw FRP context card.
- Visible state badges and patterns for observed, scaled, unknown, documented gap, and complete zero export.
- MCD64A1 context only after authentic input and independent review; otherwise keep the explicit active-fire-only state.
- Shareable evidence card with selected value, state, versions, source hashes, and limits.
- One external usability check and a public no-credential demo.

P2 — deferred until P0 and P1 pass

FIRMS Recent Pulse, NOAA-20 supplemental data, custom AOI, multilingual labels, and any new operational workflow. No spread model, evacuation recommendation, drone route, wind guidance, causal classification, or second globe belongs in this challenge path.

Score discipline

The current weighted implementation score remains **41.6 / 100**. The previous **44.8 / 100** claim ceiling and **93 / 118** local-rubric claim came from the earlier plan and are withdrawn until the perfected-plan P0/P1 gates, public URL, and eligibility ruling are complete. The perfected PDF is a blueprint, not evidence of a NASA judge score. Recalculate only from checked files, test output, or a public URL that was actually opened.

FireAtlas — Perfected Winning Plan and Accountability Tracker

Challenge: NASA Space Apps 2026 · Harmonization of MODIS and VIIRS Hot Spots **Document version:** Perfected plan alignment · 30 September 2026 **Authoritative delta:** docs/winning-plan/src/00-perfect-plan-authority.md

The supplied `FireAtlas_Updated_Perfected_Winning_Plan.pdf` supersedes the earlier plan's claim ceiling. The current weighted implementation score is **41.6 / 100**. The earlier **44.8 / 100** implementation claim and **93 / 118** local-rubric claim are withdrawn until the perfected-plan P0/P1 gates, eligibility ruling, and public demo are actually evidenced. Do not claim 54, 88, 93, 106, or 20/20 from documentation alone.

The goal

Upgrade the existing website. Keep the 3D globe and every control it has: drag to rotate, pause, zoom, reset, Satellite signals, Wildfires, the details dialog, the detections-by-day chart, and the terrain Earth link. If any of those are missing, put them back. Do not replace the globe with a diagram.

Under the globe, add the result the challenge asks for. For Northern California and for Punjab–Haryana, the calendar shows one comparable record from the years below, whether the selected season is unusual, the critical dates, and which days Suomi NPP was missing.

The perfected scientific unit is **VIIRS-equivalent active-fire cell-days on a common 1 km grid**. Native VIIRS 375 m pixels remain a detail channel. Raw FRP is shown separately in MW/day. The Sensor Bridge must make MODIS-only, VIIRS-only, both, and gap/unknown states visible without adding the two sensors into one count. These are evidence states, not a claim that the sensors have equal detection probability.

Data targets and current local archive status (do not invent rows or mark reconstructed periods complete):

Area	Box (west, south, east, north)	MODIS Collection 6.1 target start	Suomi NPP 375 m target start
Northern California	−122.2, 38.8, −120.0, 41.0	2010-07-01	2012-07-01
Punjab–Haryana	73.8, 29.5, 77.6, 32.6	2010-07-01	2012-07-01

Local status (2026-09-30): 34 sidecar-backed standard archive exports are imported. Eight newer exports retain request metadata and monthly completeness from July 2022 through June 2026 (48 MODIS and 47 S-NPP months per region; S-NPP May 2026 is unknown). Twenty-six reconstructed exports add authentic detection rows as early as July 2006 MODIS, and request 814833 adds S-NPP rows through July 2022, but their original request metadata were not supplied; they are labeled `reconstructed-rows-only`, displayed as partial detections, and never used to infer zero activity. The reconstructed MODIS inventory now includes the nominal 2019→2020 and 2021→2022 windows; the S-NPP July 2021→July 2022 window is present only as row-level evidence. The Sensor Bridge now exposes source-specific counts, `excluded_type_counts`, `evidence_state`, `coverage_state`, bridge status, and source-separated FRP for the imported paired window. It is therefore not a continuous verified historical baseline.

Download page: <https://firms.modaps.eosdis.nasa.gov/download/> · CSV · standard archive, not near-real-time. Split by year if the form requires it. No more data are necessary for the current detection-calendar demo; exact request metadata and missing archive windows are needed before the longer series can be called complete.

What "done" must look like

A judge opens the public link and, without help, sees:

1. The globe, working.
2. Two area choices: Northern California and Punjab–Haryana.
3. This sentence, filled only from the API: `{Area}, {Month Year}: {percentile}th percentile of {n} comparable years. {k} days are estimates because Suomi NPP was missing. If fewer than 10 years are comparable, the sentence is: {Area}, {Month Year}: comparison not usable. Only {n} comparable years. It must not say "unusual".`
4. The season start, peak, and end for that year.
5. A hatched day that opens the original NASA rows.
6. A method page with the held-out error of the harmonization, compared with "no harmonization" and with one single ratio.

That screen is what earns the scores in section 2. A finished code task that is not visible there does not raise the score.

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0. How to use this document

Implement the required tasks in P0/P1 order, one at a time. P2 and the operational ideas in section 7 remain deferred. The full change table and score discipline are in `src/00-perfect-plan-authority.md`.

Order	Task	Where
1	G0-T1, the human asks the Local Lead before relying on this repo at the event	Section 1
2	C2-T1, C2-T2, C2-T3, C2-T5, C2-T6, plus common-grid/FRP/state evidence	Section 4
3	C1-T1, C1-T2, C1-T3, C1-T4, C1-T6	Section 5
4	C10-T1, C10-T2	Section 5
5	C11-T1, keep the globe and put the calendar under it	Section 5
6	C8-T1, C12-T4, C12-T5, C12-T6, C12-T7	Section 6
7	P1 share card, SUB-T1, SUB-T2 (30 s), SUB-T4, and the 90 s rehearsal	Section 8

Executor system prompt

You are the executor for the FireAtlas repository (NASA Space Apps 2026).
Python 3.10+. Tests: `uv run python -m unittest discover -s tests`

RULES

1. Implement only the task id you were given. The required ids are listed in docs/winning-plan/src/01-start-here.md and the perfected-plan authority file. Do not build the replay, spread model, sector ranking, drones, wind display, chat box, extra regions, Recent Pulse, or new globe until P0/P1 pass.
2. Never remove or hide the home-page 3D globe or these controls: rotate, pause, zoom in, zoom out, reset, Satellite signals, Wildfires, details dialog, detections-by-day chart, terrain Earth link. Restore any that are missing.
3. Never invent data, dates, URLs, or results. If a download is not on disk, stop and mark the task BLOCKED.
4. Read a file before editing it. Do not refactor unrelated code.
5. Add a unit test for the new behaviour. Run the full suite. Paste the final "Ran N tests ... OK" line.
6. Hotspots are not burned area, fire perimeters, or proof of no fire. FRP is source context, not burned area or severity. Do not say the app predicts, dispatches, or recommends a response.
7. Never commit FIRMS keys or Earthdata tokens.
8. End with:
TASK: <id>
STATUS: DONE | BLOCKED (reason)
FILES CHANGED: <paths>
TESTS: <command> -> <result line>
EVIDENCE: <command output or screenshot path>
SCORECARD UPDATE: <rows>

1. Gate 0 — Eligibility (human step, before the event)

The Space Apps FAQ says teams may not begin working on the challenges before the hackathon (14–15 Nov 2026). This repo started on 27 Sep 2026. Ask the Local Lead, in writing, whether this code may be reused and whether that reuse must be disclosed. Save the reply, with email addresses removed, in `docs/winning-plan/evidence/G0-ruling.txt`. If reuse is not allowed, do not submit this repository. The scores below apply only to an eligible submission.

On 28 Oct 2026, copy the official challenge statement into `docs/winning-plan/evidence/challenge-statement-2026.md`. If it adds a requirement this plan does not cover, stop and tell the owner. Do not invent extra features to cover it.

2. The scores these tasks earn

The supplied [FireAtlas_Updated_Perfected_Winning_Plan.pdf](#) is now authoritative. Its NASA judging alignment is Impact, Creativity, Validity, Relevance, and Presentation. The weighted 100-point table below remains an internal engineering tracker, not a NASA judge score. The earlier plan's claim numbers are retained only as a historical arithmetic record and are explicitly withdrawn until the perfected P0/P1 gates pass.

$\text{points} = \text{weight} \times \text{score} \div 5$. A category stays at the "Now" score until its check passes. Do not raise a category because a diagram or a plan exists. The old targets of 54.2, 88.0, 93, and ≈ 106 are withdrawn.

2.1 Implementation score — current 41.6 / 100 · no perfected-plan claim yet

#	Category	Wt	Now	Earlier claim (withdrawn)	Points	Perfected-plan evidence gate
C1	Calendar	12	4	4	9.6	P0 archive completeness plus visible verdict, critical dates, heatmap state patterns, Sensor Bridge, and evidence drawer; older row-only history stays partial
C2	Harmonized NASA data	12	4	4	9.6	P0 common 1 km aggregation, within-sensor/day deduplication, raw 375 m detail, FRP separation, source hashes, and reproducible state classification
C3	Terrain, fuels, weather	8	1	1	1.6	No work. Do not raise
C4	Sector ranking	14	0	0	0	No work
C5	Aerial framing	7	1	1	1.4	No work
C6	Aerial observations	8	0	0	0	No work
C7	Spread scenarios	10	1	1	2.0	No work
C8	Safety wording	7	3	3	4.2	The limits sentence in section 6 is on the home page and the method page
C9	Season verdict for a manager	6	3	3	3.6	Live verdict matches the API, appears first in the calendar panel, and official sources are directly beneath it; no unsupported unusual-fire claim
C10	Held-out test	8	3	4	4.8	Held-out comparison includes common-grid, ratio/no-correction baselines, interval coverage, and the 2012 step only when complete request metadata exist
C11	Interface	4	3	4	2.4	Globe and controls remain unchanged; public URL opens the calendar and preserves See → Compare → Verify → Share
C12	Repo and demo access	4	3	4	2.4	Local static bundle plus evidence card, source ledger, hash-bound exports, and public no-credential demo
	Total	100	41.6	44.8 (withdrawn)	41.6	Current: $9.6+9.6+1.6+0+1.4+0+2.0+4.2+3.6+4.8+2.4+2.4 = 41.6$. No perfected-plan claim is set.

C1 and C2 are not claimed at 5. The perfected plan adds MCD64A1 as lagged corroboration and an outside-user check; neither is present in the current evidence bundle. A UI card cannot substitute for either input.

Perfected-plan gates (status at 30 September 2026)

Gate	Required evidence	Current status
P0 archive and provenance	Complete requested windows, metadata, hashes, UTC/type rules, and no fabricated rows	OPEN — 26 older exports are reconstructed row-only inputs; the S-NPP 2021-2022 rows are present, but their original request metadata and complete-window evidence are absent
P0 common-grid reproducibility	VIIRS 375 m aggregated to the common 1 km grid, native detail retained, deduplicated daily totals reproduced independently	PARTIAL — imported FIRMS centroid transform is documented; native mask resampling and independent review are absent
P1 Sensor Bridge and FRP	MODIS-only / VIIRS-only / both / gap states plus source-separated FRP MW/day	LOCAL UI PRESENT — counts are tied to imported exports; complete historical coverage is not established
P1 independent corroboration	MCD64A1 Collection 6.1 or separately documented official context, with dates and limits	OPEN — no MCD64A1 granules are bundled; UI says ACTIVE FIRE ONLY
P1 share card	Value, state, versions, hashes, limitations, and public URL all reproduce the selected result	LOCAL ONLY — public host not verified
P1 usability and deployment	Outside-user check, public no-credential URL, globe and mobile rehearsal	OPEN
P2 rapid mode	Separate Recent Pulse data mode, calibration, and limits	DEFERRED

2.2 Local rubric — current ≈70 / 118 · earlier 93 / 118 withdrawn

Criterion	Max	Now	Earlier claim (withdrawn)	Perfected-plan evidence gate
Impact	20	15	15	The video shows the same one-year-limited verdict in a California fire-season case and Punjab-Haryana's official 2024 paddy-harvest monitoring window , names local planners, and states hotspots do not establish crop-burning cause
Creativity	20	11	14	Visible common-grid Sensor Bridge, mismatch categories, uncertainty hatching, and retained 375 m detail
Validity	20	11	16	Held-out error, baseline comparison, interval coverage, archive hashes, and MCD64A1 context or explicit active-fire-only state
Relevance	20	11	18	The calendar is the feature the challenge names. Training lab, synthetic demo, and EONET are not on the judged path
Presentation	20	12	16	Globe remains, public link works, and both 30-second and 90-second evidence paths are rehearsed
Teamwork	5	2	2	This plan adds no teamwork tasks, so this box stays 2
User experience	5	3	4	Pick an area, read the sentence, open one hatched day, inspect the bridge, and share the evidence card; outside-user check required for 4
NASA data usage	5	4	5	Sources page lists MODIS C6.1, VIIRS S-NPP, FIRMS acknowledgement, and any MCD64A1 input with version, retrieval, and hash
Category named	1	1	1	Challenge name is on the home page
Repository access	1	0	1	Public repo matches the running site
Project page	1	0	1	Space Apps project page submitted
Total	118	≈70	93 (withdrawn)	Current: 15+11+11+11+12+2+3+4+1+0+0 = 70 ; no perfected-plan claim is set

A judge can still score a box lower if the sentence is buried or the demo fails. The earlier 93-point row is a withdrawn planning ceiling, not a prediction. Re-score only after the perfected gates are independently evidenced.

3. What to leave alone

The perfected plan explicitly retains the hero globe. Any repolish in this plan belongs below it in the calendar evidence path; it must not replace, restyle, reroute, or remove the globe, its overlays, or its controls.

The following are already out of the judged path. Do not spend another task on them: training lab, installable offline app, synthetic demo mode, EONET event feed.

Keep the 3D globe. Do not remove `globe.js`, `earth-embed.js`, `globe.css`, `terrain-earth.html`, `documented-fires.json`, or the home-page controls listed in section "The goal". Do not delete `earth.html` unless a search shows the running globe never loads it, and record that search.

Do not remove NOAA HMS code, the research page, or the recount tool. They are not part of this upgrade, and deleting them does not raise the score.

There is no further cleanup task. The next work is section 4.

4. Data and harmonization

Build this only for `norcal` and `punjab-haryana`, using the boxes and dates in section "The goal". One variant only: UTC day, FIRMS type 0 or missing (`veg`), all confidence values.

Perfected-plan measurement contract

The primary published value is **VIIRS-equivalent active-fire cell-days on a common 1 km grid**. Native VIIRS 375 m detections remain available as a detail layer. A raw MODIS pixel count and a raw VIIRS pixel count are never compared as if they were the same unit. The current FIRMS implementation assigns detection centroids to the versioned equal-area grid; it must not be described as native fire-mask resampling until standard mask products have been decoded and checked.

For every source and UTC day:

1. retain the original row, native pixel scale, product version, confidence, acquisition time, FRP, request metadata, retrieval date, file hash, and row count;
2. exclude type values outside 0 or missing from the calendar and count the exclusions;
3. deduplicate within that sensor and UTC day before aggregation;
4. aggregate VIIRS 375 m to the common 1 km cell for the shared comparison while retaining the 375 m source rows;
5. store source-specific cell-days, raw pixel counts, and raw FRP sums separately;
6. choose one reference observation for the harmonized value and expose the other sensor for comparison; never add both sensors into one activity count.

The visible quality vocabulary is `observed`, `scaled`, `unknown`, `documented_processing_gap`, and `zero_detections_exported`. A complete zero export means zero eligible rows in that export; it does not establish a clear pass, cloud-free observation, or no fire. Missing or incomplete windows remain unknown. FRP is secondary context in MW/day and is never added to the activity measure.

C2-T1 — Import the archives

- **Files:** `fireatlas/regions.py` (new), `fireatlas/archive.py`, `tests/test_archive.py`.
- **Steps:**
 1. `regions.py` holds the two regions: id, name, bbox, and one sentence on the fire regime.
 2. Import a FIRMS CSV from a folder that contains `request.json` with product (`MODIS_SP` or `VIIRS_SNPP_SP`), `start_date`, `end_date`, and `bbox`. Reject near-real-time files.
 3. A month is complete only when the request covers that whole month and the region box. Keep the existing hash and row-count checks.
 4. Remove the rule that a file must start on 1 July (`archive.py` around the July-1 check).
- **Checks:**
 - [x] Re-importing the existing Northern California 2022–2025 files still yields 30,823 rows (`tests/test_archive.py::test_packaged_nasa_pair_loads_on_fresh_database`).
 - [x] A month that sticks out of the request is stored as partial, not complete (`tests/test_archive_requests.py::test_partial_date_request_cannot_create_complete_export_window`).
 - [x] When an older CSV lacks its original FIRMS request form, its sidecar is `reconstructed-rows-only`; inferred dates never create complete export windows (`tests/test_archive_requests.py::test_reconstructed_file_dates_import_positive_rows_without_claiming_complete_coverage`).
 - [x] Missing annual windows are listed separately from supplied row-only archives in `docs/NASA_DATA_IMPORT.md`; missing dates remain unknown.

C2-T2 — Daily cell-days

- **Files:** `fireatlas/aggregates.py`, `tests/test_aggregates.py`, output `fireatlas/samples/aggregates/<region>.json.gz`.
- **Steps:** For each UTC day and each series, store cell count, raw pixel count, and FRP sum. Schema `fireatlas-daily-aggregates-v1` in the appendix. Include the input file SHA-256 and `method_version`.
- **Checks:**
 - [x] For Northern California, all 31 July 2024 MODIS daily cell counts match the existing calendar using the authentic bundled NASA rows (`tests/test_archive.py::test_packaged_nasa_pair_loads_on_fresh_database`).
 - [x] A row with FIRMS type 2 is excluded (`tests/test_aggregates.py::test_daily_aggregate_excludes_type_two_and_preserves_zero_day`).

C2-T3 — Missing-satellite days

- **Files:** `fireatlas/availability.py`, `fireatlas/samples/sensor_notices.json`, `tests/test_availability.py`.
- **Steps:**
 1. Seed `sensor_notices.json` with a dated, sourced S-NPP product notice. The verified 2024 interval is 2024-07-24 05:24 UTC through 2024-07-29 15:18 UTC, inclusive. The source labels a product-processing outage; it does not establish a no-fire condition or cloud mask. Preserve the exact interval, source URL, retrieval date, and a quote of 25 words or fewer.
 2. Never infer satellite unavailability from a zero FIRMS count. A complete export with zero rows is `zero_detections_exported`; its pass, cloud and observation opportunity remain unknown. An incomplete export is

- unknown_export. A source notice intersecting a day is documented_processing_gap, even if the export contains some points.
3. Remove hardcoded Park Fire date checks from app.js. A date hatch and explanation must come from the source-notice ledger; no-pass and cloud states stay unknown unless native coverage inputs establish them.

- **Checks:**

- [x] 2024-07-25 in Northern California is flagged documented_processing_gap for Suomi NPP; the sourced notice ID and URL are retained (tests/test_validity.py::test_real_calendar_api_and_day_drawer_keep_nasa_source_evidence, tests/test_aggregates.py::test_zero_exports_and_notices_are_not_called_sensor_pass_or_missing_fire).
- [x] A complete day with zero Suomi NPP rows is zero_detections_exported, with observation opportunity unknown (tests/test_aggregates.py::test_zero_exports_and_notices_are_not_called_sensor_pass_or_missing_fire).
- [x] A complete day with Suomi NPP rows and no notice is detections_in_export, with observation opportunity still unknown (tests/test_aggregates.py::test_zero_exports_and_notices_are_not_called_sensor_pass_or_missing_fire).

C2-T5 — The ratio that joins the sensors

- **Files:** fireatlas/calibration.py, tests/test_calibration.py, fireatlas/samples/calibration/<region>.json.

- **Method, implement exactly:**

1. Overlap months are complete months from 2012-07 through the last month both products have, skipping only dates intersecting a documented S-NPP product-processing gap. Complete zero-detection exports remain observed zeros.
2. For month-of-year m , $r(m) = (\text{sum of Suomi NPP cell-days}) / (\text{sum of MODIS cell-days})$ over those overlap years. If the MODIS sum for that month is under 30, use the all-year ratio instead and set basis to annual-fallback.
3. Uncertainty: resample overlap years with replacement 1,000 times using `random.Random(20261114)`. The 95% interval is the 2.5th and 97.5th percentiles.
4. For dates before the downloaded S-NPP archive starts (2024 plan: 2012-07-01), and on dates intersecting a documented S-NPP product outage: harmonized value = MODIS cell-days $\times r(m)$, estimate_type = scaled, source_used = MODIS_SP. A complete S-NPP export with zero eligible records is an observed zero-detection record, not a missing sensor. An incomplete S-NPP export outside a documented outage is unknown, not an estimate.
5. Leave-one-year-out: refit without each overlap year, predict that year's Suomi NPP monthly totals, and record median absolute log error `median(|ln((predicted+1)/(actual+1))|)`, annual absolute percent error, and how often the actual falls inside the interval. Compute the same error for $r = 1$ and for one all-year ratio.
6. The calendar uses whichever of the three has the lower leave-one-year-out log error. Say which one on the method page.

- **Checks:**

- [x] Synthetic test where Suomi NPP = 4 $\times$ MODIS: $r(m)$ is 4 and held-out error is 0 (tests/test_calibration.py::test_fixed_ratio_has_zero_held_out_error_and_reproducible_seed).
- [x] Both regions have calibration JSON artifacts; the web test loads and schema-checks each (tests/test_web.py::test_calibration_artifacts_are_downloadable_and_method_page_uses_them).
- [x] Method-page values are loaded from the selected calibration JSON; the HTML contains placeholders and the script reads model metrics from the JSON.

Perfected validation additions

The ratio is not a license to call the sensors equally sensitive. Before a perfected-plan claim, the reviewer must also check:

- daily and monthly common-grid totals against the retained source rows after deduplication;
- MODIS-only, VIIRS-only, both, gap, and unknown categories on complete and incomplete days;
- raw FRP sums per source against the same filtered rows, without cross-sensor addition;
- sensitivity to the common-grid rule and to confidence/type filters;
- leave-one-year-out performance against no correction, one annual ratio, and the selected model, with interval coverage shown even when it is weak;
- MCD64A1 Collection 6.1 only as lagged burned-area context, never active-fire ground truth.

The current repository has no MCD64A1 granules and no independent sign-off. Until those inputs exist, the method page must show ACTIVE FIRE ONLY and a NASA product link rather than a blank burned-area overlay or a made-up corroboration value.

C2-T6 — Sources

Add docs/DATA.md rows for both products: name, version, URL, date range, retrieval date, licence, SHA-256 of the imported files. Use this FIRMS acknowledgement, and re-check it before submission: "We acknowledge the use of imagery from the NASA LANCE FIRMS (<https://earthdata.nasa.gov/firms>), part of the NASA Earth Science Data and Information System (ESDIS)."

Master prompt: TASK C2-T<n>. Follow docs/winning-plan/src/04-gate1-and-data.md for that task only. Use the method as written. Output the REPORT block.

5. Calendar, proof, and the home page

Perfected-plan judge path

Keep the existing globe and controls at the top. Under it, follow **See → Compare → Verify → Share**. The calendar must communicate the result before the user opens the method text:

1. **Verdict:** percentile/comparable years, estimated days, unknown days, and official links.
2. **Sensor Bridge:** MODIS raw source values → common 1 km transformation → VIIRS-equivalent value, with MODIS-only, VIIRS-only, both, and gap/unknown states visible.
3. **Heatmap:** years by UTC day, with observed solid, scaled amber hatch, unknown crosshatch, documented gap purple hatch, and complete zero export labelled. Every pattern has a text label and keyboard-readable description.
4. **Evidence drawer:** source values, native scales, FRP in MW/day per sensor, quality state, notice, source rows, version, hash, and a download action.
5. **Share card:** selected value, state, versions, source inputs/hashes, limits, and a URL generated from the same API/static JSON. It is a briefing artifact, not a new result.

The MCD64A1 panel is shown only when authentic dated granules and an independent review record exist. Otherwise the page says **ACTIVE FIRE ONLY** and links to NASA's product record. Do not show a burned-area value, fire perimeter, fire-free area, or validity percentage based on the absence of that product.

C1-T1 — Calendar API

- **File:** `fireatlas/calendar_v2.py`. Route `GET /api/v2/calendar`.
- **Parameters:** `region` (`norcal` or `punjab-haryana` only), `year`. Reject anything else with HTTP 400.
- **Response:** schema `fireatlas-calendar-v2` in the appendix. For each day: `harmonized value`, `low`, `high`, `estimate_type`, `source_used`, `quality` (`good` for complete S-NPP exports, `degraded` for a MODIS estimate outside the downloaded S-NPP window or within a documented product gap, `unknown` for incomplete exports or unvalidated scaling), `evidence_state`, `coverage_state`, `sensor_bridge`, and source-separated `frp_mw_day`. A scaled value during a documented gap has `evidence_state=scaled` and `coverage_state=documented_processing_gap`; the UI says "gap · scaled estimate". A complete export with zero records is an observed zero-detection day; pass/cloud coverage stays unknown. For each month: `value`, percentile among baseline years, `rank`, `n_years`, and `flag` (`unusually-high` at or above the 90th percentile, `unusually-low` at or below the 10th, `typical`, or `insufficient-history` when `n_years < 10`).
- **Baseline:** years from 2010 through the year before the selected year, and only years whose MODIS collection matches the selected year. List excluded years.
- **Checks:**
 - [x] Northern California 2024 returns 12 months from authentic bundled NASA records through `/api/v2/calendar` (`tests/test_validity.py::test_real_calendar_api_and_day_drawer_keep_nasa_source_evidence`).
 - [x] A third region id returns 400 (`tests/test_web.py::test_region_calendar_api_is_named_scoped_and_never_fabricates_empty_data`).

C1-T2 — Heatmap

Under the globe, years as rows and days as columns, 2010 through the latest complete year. An incomplete export, a documented product-gap interval, and a complete zero-detection export have distinct patterns and labels. A daily outage hatch means the UTC day intersects the notice; it does not mean the whole day had no S-NPP data. Hover or keyboard focus reads the date, the value, the unit "VIIRS-equivalent active-fire cell-days", and whether the value is observed, scaled, unknown, a documented processing gap, or a complete zero export.

- [x] Screenshot `docs/winning-plan/evidence/C1-T2-norcal.png` shows the 2010s, unknown export months, and the documented S-NPP outage interval intersecting 24–29 July 2024 with partial endpoint days marked distinctly. Captured from the live local archive on 2026-09-29 at 1440 CSS px / DPR 2; includes the July 2024 daily grid and full historical legend.

C1-T3 — The verdict sentence

First text in the calendar panel. Templates, filled only from the API:

- `{area}, {month} {year}: {percentile}th percentile of {n} comparable years. {k} days are MODIS estimates before the downloaded S-NPP period or during a documented product gap.`
- If `n < 10`: `{area}, {month} {year}: comparison not usable. Only {n} comparable years.`

Under the sentence, two links:

- Northern California: CAL FIRE incidents <https://www.fire.ca.gov/incidents> and InciWeb <https://inciweb.wildfire.gov>
- Punjab-Haryana: NASA FIRMS, plus a state source only if you loaded the page and got HTTP 200. Otherwise the line is `No verified official local source listed.`
- [x] `tests/test_calendar_v2.py::test_comparison_verdict_names_estimate_reason_and_omits_zero_estimate_clause` covers the percentile and insufficient-history templates, names the two MODIS-estimate reasons, and omits the estimate clause at zero.

- [x] Live API and browser check for July 2024, Northern California: the visible verdict exactly matches `meta.verdict`; CAL FIRE and InciWeb links sit directly beneath it. Screenshot: `docs/winning-plan/evidence/C9-T1-verdict.png`.

C1-T4 — Critical dates

For each year, using harmonized daily values and skipping missing months: season start is the first day whose cumulative share reaches 10%, season end is the first day it reaches 90%, and the peak is the centre of the 15-day window with the largest sum. If more than 10% of days are missing, season status is `unavailable`. Show start, peak, and end next to the verdict.

- [x] A synthetic triangle-shaped year returns the known start, peak, and end
(`tests/test_calendar_v2.py::test_triangle_returns_known_ten_percent_dates_and_fifteen_day_peak`).

C1-T6 — Day drawer

Clicking a day opens the existing evidence drawer: the harmonized value, each sensor's availability, and up to 200 original rows. The button label remains a download of that day's CSV.

- [x] 25 July 2024 returns original MODIS source rows and a documented S-NPP processing gap; the drawer links to the NASA notice from the API and does not treat missing rows as proof of no fire or no pass
(`tests/test_validity.py::test_real_calendar_api_and_day_drawer_keep_nasa_source_evidence`,
`tests/test_aggregates.py::test_zero_exports_and_notices_are_not_called_sensor_pass_or_missing_fire`,
`tests/test_web.py::test_calendar_map_observations_and_exports`).

C10-T1 and C10-T2 — Method page

On `/method.html`, show one section, "Does the scale hold up?":

- The leave-one-year-out model comparison from the calibration JSON: monthly-ratio error, single-ratio error, no-harmonization error, and which one the app uses. The existing horizontal comparison graphic is the visual presentation of those values.
- One line for the 2012 step: the 2012 annual Suomi NPP / MODIS ratio, and whether it sits inside the 95% interval fitted on the other overlap years.

The perfected plan adds the compact Sensor Bridge and FRP audit to the existing method comparison; it does not add a second globe, a spread forecast, or an operational recommendation. Any MCD64A1 context remains a lagged corroboration note with its own source and date.

- [x] The model errors, annual errors, selected model, and interval-coverage values displayed for both regions match the corresponding calibration JSON (`uv run --with playwright python scripts/verify_calibration_ui.py --base http://127.0.0.1:8765`; captures: `docs/winning-plan/evidence/C10-T1-calibration-norcal.png` and `C10-T1-calibration-punjab-haryana.png`).
- ☐ The 2012 step is still pending because row-only legacy imports do not establish complete monthly source exports; do not show a ratio as a measured complete-window result until that input gate passes.

C11-T1 — Home page

Order, top to bottom: header, **existing globe with all of its controls**, verdict sentence and official links, Sensor Bridge, daily heatmap/calendar, critical dates and method context, source-row drawer, share card, link to the method page, and footer with the limits sentence from section 6. Add the challenge name Harmonization of MODIS and VIIRS Hot Spots in the eyebrow. Do not remove or edit globe controls to make room.

Master prompt: TASK `<id>`. Follow `docs/winning-plan/src/05-calendar-validation.md` for that task only. Numbers on screen come from the API. Output the REPORT block.

6. Publish what the judge will open

The perfected release path is a no-credential evidence demo. The public copy must carry the same static JSON used by the calendar, Sensor Bridge, method audit, day drawer, and share card. It must not silently fall back to localhost or a synthetic fixture.

C8-T1 — Limits sentence

Put this exact sentence in the home-page footer and on `/method.html`:

FireAtlas is a research and learning tool. It is not an operational fire-management, evacuation, or flight-planning tool.

- [x] Both pages contain that sentence (`tests/test_web.py::test_winning_plan_safety_and_data_disclosures`).

C12-T4 — Licence

Put the Apache License 2.0 text from <https://www.apache.org/licenses/LICENSE-2.0.txt> into `LICENSE` without retyping it. State in the README that NASA data is not covered by that code licence.

C12-T5 — README

Rewrite `README.md` to at most 120 lines: what the site does, the public link, how to run it (`uv sync` then `bash scripts/launch_demo.sh`), the two regions, the data versions, and the limits sentence. Say that the home page opens on the 3D globe.

C12-T6 — Public site

`scripts/export_static.py` writes a `site/` folder that contains the globe assets and pre-built `/api/v2/calendar` JSON for both regions and each year from 2010 through the latest complete year. The calendar must render from those files when the page is opened as static files. Deploy that folder to GitHub Pages.

The export also includes dated Park/Grove validity reports, analytical recount results, evidence ZIPs, standalone blank native-review forms, and a read-only Data Sources ledger. Run `scripts/verify_static_calendar.py`, `scripts/verify_static_method.py` for both cases, and `scripts/verify_static_data.py` against the release folder; these checks must pass without `/api/` requests. The static copy disables CSV imports and NASA sync controls while preserving the dated provenance and limitations.

- ☐ The public URL loads both regions with no request to `127.0.0.1`.
- ☐ The globe still renders on that URL.

Perfected-plan release checks

- The header names the challenge and current data status.
- The Sensor Bridge labels raw MODIS, common 1 km transformation, VIIRS-equivalent activity, and native 375 m detail without adding sensor counts.
- FRP is shown as source-separated MW/day context, never as the calendar unit.
- State badges and patterns distinguish observed, scaled, unknown, documented processing gap, and complete zero export.
- The shareable evidence card includes selected value, state, product versions, source hashes, limitations, and the public URL. Its values are generated from the selected JSON, not baked into an illustration.
- If MCD64A1 is unavailable, the page says `ACTIVE FIRE ONLY` and links the NASA product. A release must never imply burned-area corroboration exists when the granules are absent.
- Rapid Pulse, if added later, is a separate mode with its own freshness, calibration, and limits; it cannot change historical calendar values.

C12-T7 — Sources page

`/data.html` lists MODIS Collection 6.1 and VIIRS Suomi NPP 375 m with version, URL, date range, retrieval date, file SHA-256, and the FIRMS acknowledgement from C2-T6. If MCD64A1 is acquired, list its Collection 6.1 DOI and retrieval/hash details beside its lagged corroboration role. Link `docs/AI_USE.md`, which names the tools used, says they did not generate data values, and names the people who reviewed the numbers. Until a reviewer signs off, say that independent review is pending.

Master prompt: TASK <id>. Follow `docs/winning-plan/src/06-packaging-and-safety.md` for that task only. Output the REPORT block.

7. Do not build this in the challenge path

Do not build a fire-spread replay, terrain analysis, fuels, weather, wind display, sector ranking, drone or airspace guidance, chat with the data, or a "most effective response" view.

Those features are not required by the challenge statement, they are not in the score claimed in section 2, and a model that builds them instead of section 4 and section 5 will miss the claimed score. If a pull request adds them, reject it.

The perfected plan also defers FIRMS Recent Pulse, NOAA-20 supplemental data, custom AOI, and multilingual labels until P0/P1 are complete. If a rapid product is added later, it must be a separate data mode with its own freshness, calibration, and limits; it may not alter the historical calendar. The existing globe remains the only globe.

8. Submission

SUB-T1 — Project page

Fill the Space Apps project page with: the challenge name, the public URL, the public repo, a 150-word summary that states the verdict and comparable-year count for both regions using numbers copied from the method page, the NASA products and versions, the common-grid unit, the separate FRP context, the limits sentence, and AI use from `docs/AI_USE.md`. If a number is not in `SCORECARD.md` or the current API, write `[MISSING]` rather than guessing. If `MCD64A1` is not present, say `ACTIVE FIRE ONLY` rather than implying corroboration.

SUB-T2 — 30-second video

Seconds	On screen	Words
0-4	The globe	"MODIS and VIIRS both detect fires, but their counts are not the same record."
4-12	Northern California heatmap and the verdict sentence	Read the sentence exactly as the page shows it.
12-18	A hatched July 2024 day and its NASA rows	"Days a satellite was missing stay marked."
18-24	Punjab-Haryana · October 2024	"Local planners can inspect detections during the official harvest-monitoring window; one comparable year is not enough for a trend claim, and hotspots do not identify cause."
24-30	Method-page error line, then the URL	"On years we held out, the error is {the figure on the page}."

The perfected plan also requires a separate 90-second narrated rehearsal:

Seconds	On screen	Narration
0-5	Existing globe	"The record changes when the sensor changes."
5-15	Sensor Bridge	"Raw MODIS and VIIRS pixels are different units; the calendar uses a common 1 km grid."
15-35	Northern California calendar	Read the live verdict and one state badge exactly as shown.
35-50	Evidence drawer	Show source values, native scale, FRP per source, notice, version, and hash.
50-65	Punjab-Haryana calendar	Explain the official context and that hotspots do not establish cause.
65-80	Method page	Show held-out errors, baseline comparison, and interval coverage.
80-90	Share card	Show the selected value, state, sources, limits, and share URL.

The 90-second sequence is a target until it is captured from the current build and checked on the public URL. The local silent 30-second cut is evidence of the earlier path only; do not describe it as proof that the perfected sequence has been rehearsed.

SUB-T4 — Demo

Open the public URL. Show the globe for five seconds. Switch to Northern California, read the verdict, open one hatched day. Switch to Punjab-Haryana and read that verdict. Open the method page and read one error number. Stop. Do not open a spread model, a wind map, or a chat box.

9. How to confirm the score

After SUB-T4 and the perfected 90-second sequence have been rehearsed on the public URL, update the SCORECARD only if every P0/P1 check in sections 4–6 is ticked. The old 44.8 and 93 figures are withdrawn; calculate any new value from the evidence actually available. If a gate is open, leave that category at its current score. Do not average, round up, or copy a planning target.

Auditor prompt:

Open `docs/winning-plan/SCORECARD.md` and section 2 of `docs/winning-plan/src/02-gates-and-scoring.md`. For each perfected gate, pass the check only if the evidence column cites a command output, a file, or a public URL you opened. Keep the current score while a gate is open; calculate a new score only after all required P0/P1 evidence is present and the auditor records the arithmetic. Do not raise a score for work outside sections 4–8 or for an unbundled MCD64A1 claim.

Appendix — Fields the tasks must use

Dates are YYYY-MM-DD and timestamps are UTC. Missing values are `null`. The perfected plan adds the fields below so a judge can distinguish a source observation from an estimate or an export gap. Do not infer pass, cloud, no-pass, burned area, or fire-free status from a missing row.

Aggregates (`fireatlas-daily-aggregates-v1`)

Top level: `schema`, `method_version`, `region` (`id`, `bbox`), `generated_utc`, `inputs` (file name, request metadata, retrieval date, and `sha256`), `days`. Each day: `date`, and for MODIS_SP and VIIRS_SNPP_SP: `cells`, `raw`, `frp_sum`, `excluded`, and `excluded_type_counts`, plus a `sensor_bridge` object with its method version, common-grid counts and mismatch state. The full daily aggregate response additionally records native pixel size, the centroid-to-EASE-Grid-1km method, eligible and excluded row counts, FRP valid/missing rows, product versions, and source hashes. Keep VIIRS 375 m detail rows separately inspectable. A documented outage can leave FRP unknown; it is never silently shown as zero.

Calibration (`fireatlas-calibration-v1`)

Top level: `schema`, `region`, `overlap` start and end, `bootstrap` replicates 1000 and seed 20261114, `ratio_by_month` (`month`, `r`, `r_lo`, `r_hi`, `basis`), `lovo` with `month_of_year`, `single_ratio`, and `no_harmonization` (each has `median_abs_log_error`), and `selected_model`.

Calendar (`fireatlas-calendar-v2`)

`meta`: `region`, `bbox`, `unit` = VIIRS-equivalent active-fire cell-days on a common 1 km grid, `native_detail`, `bridge_method_version`, `frp_context`, `corroboration`, `baseline` (`years_used`, `excluded`), `calibration_id`, `data_class`, and source inputs/hashes. `days`: `date`, `value`, `low`, `high`, `estimate_type` (observed OR scaled), `source_used`, `quality`, `evidence_state` (observed, scaled, unknown, OR `complete_zero_export`), `coverage_state` (observed, unknown, `documented_processing_gap`, OR `complete_zero_export`), `sensor_bridge`, `frp_mw_day`, source-specific raw cells/pixels/FRP, and the reason. A scaled outage day therefore carries both `evidence_state=scaled` and `coverage_state=documented_processing_gap`. `months`: `month`, `value`, `percentile_rank`, `rank`, `n_years`, `flag`, `degraded_days`, `unknown_days`, `season_start`, `season_peak`, `season_end`, `season_status`.

Notices

`id`, `source_id`, `start_utc`, `end_utc`, `url`, `retrieved_utc`, `quote`, and the affected product/version. A notice marks a documented processing gap; it does not prove no pass or no fire.

Appendix D — SCORECARD snapshot at build time

The live version is `docs/winning-plan/SCORECARD.md`; rebuild this PDF to refresh the snapshot.

FireAtlas SCORECARD — live accountability tracker

Plan: `docs/winning-plan/FireAtlas_Updated_Perfected_Winning_Plan.pdf` (authoritative, 30 September 2026) **Historical snapshot:** `docs/winning-plan/FireAtlas_Winning_Plan.pdf` version 1.2 **Rule:** Only evidence can change the score. The current weighted implementation score is **41.6 / 100**. The earlier **44.8 / 100** and **93 / 118** ceilings are withdrawn while the perfected P0/P1 gates remain open. Do not use 54.2, 88, 93, 106, or 20/20 as a promised result.

Field	Value
Last audit	2026-09-30
Implementation score	41.6 now · no perfected-plan claim while P0/P1 gates are open
Local rubric	≈70 now · earlier 93 ceiling withdrawn
Branch	r7-work
Public repo URL	GitHub origin configured; public access not verified
Public demo URL	Not deployed

Perfected-plan delta

The perfected plan changes the evidence contract without changing the globe. The primary unit is **VIIRS-equivalent active-fire cell-days on a common 1 km grid**; native VIIRS 375 m detail and raw FRP in MW/day remain separate. The visible path is **See → Compare → Verify → Share**. MCD64A1 Collection 6.1 is a lagged corroboration option, not active-fire ground truth. When it is unavailable, the app must say **ACTIVE FIRE ONLY**. See [src/00-perfect-plan-authority.md](#) for the complete delta, state vocabulary, P0/P1/P2 order, and score discipline.

Perfected gate	Current state	Score treatment
P0 archive/provenance/common-grid reproducibility	Partial: 26 older exports are reconstructed; the S-NPP 2021–2022 rows are now present as row-only evidence but lack original request metadata; native mask resampling, full-footprint accounting and independent review are absent	No uplift
P1 Sensor Bridge and raw FRP	Local UI present for imported source exports; source counts and FRP remain separate	No uplift until release checks pass
P1 MCD64A1 context	No MCD64A1 granules in the bundle; UI states ACTIVE FIRE ONLY	No corroboration claim
P1 share card and public demo	Local card/static bundle verified; public host not verified	No uplift
P1 outside-user check and 90-second rehearsal	Local 90-second rehearsal verified; outside-user check not recorded	No uplift
P2 Recent Pulse/NOAA-20/custom AOI	Deferred	Do not build before P0/P1

Required tasks

ID	Task	Status	Evidence
G0-T1	Written reuse ruling	BLOCKED	No ruling is recorded. A copy-ready request and the current official-rule evidence are in <code>docs/winning-plan/evidence/G0-ruling-request.md</code> ; the participant must send it to their own Local Lead and save the response.
C2-T1	Import historical MODIS and Suomi NPP for both regions	PARTIAL	Imported 34 sidcar-backed standard CSVs: eight with saved request metadata and 26 older CSVs with locally reconstructed row-only sidecars. The database contains 1,994,162 standard rows (456,202 MODIS and 1,537,960 S-NPP); six MODIS archives add 154,660 rows from July 2006 onward, and S-NPP request 814833 adds 204,694 unique clipped rows dated July 2021–July 2022. The S-NPP rows are now present, but their original request metadata and all reconstructed-window completeness remain absent; older absence stays unknown.
C2-T2	Daily cell-days, common-grid bridge, FRP and exclusions	DONE / PARTIAL perfected gate	<code>fireatlas/samples/aggregates/{norcal,punjab-haryana}.json.gz</code> contain 1,430 paired-complete UTC days per region with parent hashes. The bundles now expose <code>excluded_type_counts</code> , <code>per-source FRP</code> , and <code>sensor_bridge</code> status/categories; the calendar exposes <code>evidence_state</code> . Northern California July 2024 MODIS counts match the existing calendar on all 31 days in the authentic bundled archive; type-2 exclusion and explicit zero rows are tested. Grove native evidence now decodes 20/20 expected granules into 15,495 clipped pixels and reconciles 7/7 available FIRMS rows; Park decodes 114/114 into 1,372,872 pixels and reconciles 3,137/3,137 rows. Both native queues remain processed-unreviewed; full-footprint pass/cloud coverage remains absent.
C2-T3	Product-outage ledger	DONE	The sourced 2024 S-NPP outage interval and short quote are in <code>fireatlas/samples/sensor_notices.json</code> ; unit tests cover partial endpoints, affected days, complete zero exports, and unknown opportunity. Calendar gap labels come from the ledger, not hard-coded fire dates.

ID	Task	Status	Evidence
C2-T5	Ratio, uncertainty, leave-one-year-out	PARTIAL	fireatlas/samples/calibration/{norcal,punjab-haryana}.json bind the 41 eligible 2023-2026 MODIS 61.03 / VIIRS 2 month-pairs to parent CSV and daily-bundle hashes. Held-out interval coverage is only 41.5% / 39.0%; calibration is weak, 2012 remains pending, and independent review is open. The perfected common-grid/bridge method is now visible, but no MCD64A1 corroboration is present. MODIS 6.03 is not mixed into this fit.
C2-T6	Sources and FIRMS acknowledgement	PARTIAL	docs/DATA.md and Data Sources page document the supplied parent hashes, products, limits and acknowledgement; pre-2022 files and retrieval dates are unavailable.
C1-T1	Region-aware calendar API	PARTIAL	An integration test calls /api/v2/calendar over the authentic bundled archive and verifies all 12 Northern California 2024 months. The response now carries evidence_state, bridge status, source-specific values, and unknown/gap branches. The larger local imports reach positive MODIS detections from July 2006 and S-NPP detections through July 2022; those older inputs are row-only and cannot support a complete harmonized monthly baseline.
C1-T2	Calendar with source/gap states	DONE / PARTIAL perfected gate	Live headless-browser review confirmed 2006-2026 history, partial and unknown older archive windows, July 2024 gap dates 24-29, distinct partial-day classes on 24 and 29, and keyboard-readable date/value/source labels. evidence_state is visible for observed, scaled, documented processing gap, complete zero export, and unknown branches. The replacement evidence capture is docs/winning-plan/evidence/C1-T2-norcal.png (2880x2800); native pass/cloud/no-pass coverage is still unknown.
C1-T3	Verdict sentence and official limits	DONE	Unit tests cover the percentile and insufficient-history branches, estimate reasons, and zero-estimate omission. The live July 2024 Northern California UI verdict matches the API exactly; CAL FIRE and InciWeb links are immediately below it. Capture: docs/winning-plan/evidence/C9-T1-verdict.png.
C1-T4	Season start, peak, end	DONE	Synthetic triangle returns known 10% start, 15-day peak center, and 90% end; >10% missing days withhold all dates. Real 2023 API results are available with zero missing days for both regions and are displayed in the critical-dates card.
C1-T6	Day drawer with original rows	DONE	A server-backed test over the authentic bundled archive verifies that 25 July returns original MODIS rows and a documented S-NPP processing gap. The drawer now renders an explicit NASA notice link from the API. Tests verify the archive rows, notice URL, and link-rendering path.
C10-T1	Held-out error	DONE / PARTIAL perfected gate	scripts/verify_calibration_ui.py compares each region's displayed model errors, annual errors, selected method, interval coverage, and 2012 step state with its source JSON; it asserts a pending step shows no ratio/result marker. Captures: docs/winning-plan/evidence/C10-T1-calibration-*.png. Common-grid bridge and FRP display are source-linked, but C10-T2, complete 2012 metadata, native mask review, and independent corroboration remain open; C10 stays at 3.
C10-T2	2012 step test on the method page	PARTIAL	July 2012 S-NPP rows are loaded, and the 2021-2022 row archive is now present, but the original export request metadata are missing and the legacy calendar marks these periods partial. A coverage-valid sensor-step test is not justified yet.
C11-T1	Globe kept; calendar under it	PARTIAL	The app and static embed scripts currently point to /terrain-earth.html?embed=landing, and method-page calendar links return to the home calendar. Route and JavaScript checks pass; a fresh Chrome run found the frame with no page errors but the ArcGIS terrain renderer did not reach its ready state within 15 seconds, so Earth appearance and overlays remain unverified. No public demo URL is verified; C11 stays at 3.
C8-T1	Limits sentence on home and method	DONE	Exact required text is in the visible home and method footers; full tests and desktop/mobile page checks passed.
C12-T4	Apache-2.0 licence	DONE	Root LICENSE uses the official Apache 2.0 text with the project copyright field filled. NASA data are excluded.
C12-T5	README	DONE	Root README.md is 35 lines and describes the current app, actual sample coverage, setup and limits.
C12-T6	Public static site	PARTIAL	Regenerated site/ with 42 calendars (2006-2026), 113 manifest-listed checksummed data files, two history files, compressed daily source rows, dated validity files, two evidence ZIPs, standalone hash-bound blank native-review templates for both cases, and Sensor Bridge/share-card data. Static calendar, Data & Method, and Data Sources browser verifiers run without /api/ requests; server-only imports stay disabled. Public deployment and globe behavior remain unverified.
C12-T7	Sources page and AI use	PARTIAL	Data page and docs/AI_USE.md state AI use and disclose that no independent numerical reviewer is recorded; sign-off remains open.
SUB-T1	Project page	DRAFT READY — NOT SUBMITTED	Copy-ready, exactly 150-word draft is in docs/winning-plan/SUBMISSION_DRAFT.md. Public URLs are missing, submission is external, and G0 is unresolved.
SUB-T2	30-second video + perfected 90-second story	LOCAL 30 s + 90 s REHEARSAL VERIFIED — PUBLIC CAPTURE OPEN	docs/winning-plan/evidence/FireAtlas_30s_demo.mp4 and docs/winning-plan/evidence/FireAtlas_90s_perfected_demo.mp4 are verified silent H.264 cuts (1600x900, 30 fps). The 90-second sequence uses the same local static bundle for the unchanged globe frame, Sensor Bridge, source-separated FRP, evidence drawer, Grove native-mask audit (20/20 granules and 7/7 available FIRMS rows matched), Park native-mask audit (114/114 granules and 3,137/3,137 FIRMS rows matched), Punjab context, and share card; scripts/render_perfected_plan_video.py checks the stream, duration, frame hashes, static-bundle hash and WebVTT sidecar. The Punjab scene shows October 2024's 5,174 detections, one comparable year, and the dated government monitoring context with an explicit no-cause-attribution caveat. Public URL, voice-over, deployment, and submission remain open; the local captions are a timed sidecar, not a public capture. The opening Earth is a still capture and was not freshly rendered.
SUB-T4	Demo rehearsed on the public URL	NOT STARTED	

Anything not in this table is out of the plan. Do not implement it.

Category scores

Current implementation arithmetic uses the plan's $\text{weight} \times \text{score} \div 5$ rule. The verdict now appears before the calendar heading, with the official links directly beneath it, and the live Northern California July 2024 text matches the API. C9 moves from 2 to 3. The perfected plan adds visible `evidence_state`, `sensor_bridge`, `excluded_type_counts`, source-separated FRP, and the share card; these artifacts improve inspectability but do not raise a score until their P0/P1 release gates pass. C1 remains 4 for the verified 2022–2026 window; older row-only dates remain partial. C10 is 3; the longer-history baseline, 2012 step test, MCD64A1 context, and independent review remain unverified.

The local database includes the supplied pre-2022 source rows in both regions, including the reconstructed S-NPP 2021–2022 archive. The implementation score is **41.6 / 100** under this plan. The calendar verdict is now first, source-linked, and verified against the live July 2024 API result. C1 meets the plan's real-data calendar gate for the verified historical window, and C2-T2 passes for the paired 2022–2026 window. The static calendar is locally reproducible, but static globe metadata and observation requests still fail, and no public URL is deployed. This does not establish a complete 2010/2012 baseline: 26 older CSVs lack original request metadata, the S-NPP 2021–2022 rows are reconstructed rather than request-verified, and independent validation remains open. This is the plan's local estimate, not a NASA judge score.

#	Weight	Current score	Current points	Earlier claim (withdrawn)	Earlier claim points	Perfected-plan evidence gate
C1	12	4	9.6	4	9.6	P0 complete windows plus verdict, critical dates, heatmap patterns, Sensor Bridge, and drawer
C2	12	4	9.6	4	9.6	P0 common 1 km unit, deduplicated source totals, native detail, excluded-type counts, FRP, and bridge states
C3	8	1	1.6	1	1.6	Unchanged
C4	14	0	0	0	0	Unchanged
C5	7	1	1.4	1	1.4	Unchanged
C6	8	0	0	0	0	Unchanged
C7	10	1	2.0	1	2.0	Unchanged
C8	7	3	4.2	3	4.2	Exact limits sentence visible on home and method pages
C9	6	3	3.6	3	3.6	Live verdict matches the API, appears first, and has official sources immediately beneath it
C10	8	3	4.8	4	6.4	Common-grid held-out comparison, interval coverage, baseline comparison, and complete 2012 step
C11	4	3	2.4	4	3.2	Globe and controls unchanged; public URL preserves See → Compare → Verify → Share
C12	4	3	2.4	4	3.2	Static evidence bundle, share card, source ledger, hashes, and public no-credential demo
Total	100		41.6	44.8 (withdrawn)	44.8	Current: 9.6+9.6+1.6+0+1.4+0+2.0+4.2+3.6+4.8+2.4+2.4 = 41.6 ; no perfected-plan claim is set

Local criterion	Earlier claim (withdrawn)
Impact	15
Creativity	14
Validity	16
Relevance	18
Presentation	16
Teamwork	2
User experience	4
NASA data usage	5
Category named	1
Repository access	1
Project page	1
Total	93 (withdrawn)

Data on disk today

Region	Product	Years	Rows
norcal	MODIS_SP	positive rows from 2006-07; verified complete months 2022-07 through 2026-06	54,659
norcal	VIIRS_SNPP_SP	rows from 2012-07 through 2022-07 are reconstructed; verified complete months 2022-07 through 2026-06 except May 2026	173,245
punjab-haryana	MODIS_SP	positive rows from 2006-07; verified complete months 2022-07 through 2026-06	401,543

Region	Product	Years	Rows
punjab-haryana	VIIRS_SNPP_SP	rows from 2012-07 through 2022-07 are reconstructed; verified complete months 2022-07 through 2026-06 except May 2026	1,364,715

The 26 older CSVs have reconstructed date-only sidecars and are never counted as complete months. Six newly supplied MODIS files add 154,660 positive rows, including July 2006–2010 history and nominal 2019–2022 windows; S-NPP request 814833 adds 204,694 unique clipped rows for July 2021–July 2022. On the live site, use the Data Sources ledger to distinguish them from the 2022–2026 request-verified exports. A clean clone does not include the ignored 28 GB `NASA_data/` archive or this local database.

Park Fire, 25 July 2024: the validity view has MODIS 603 raw pixels / 415 centroid cells and Suomi NPP 0 imported rows. The calendar estimates 925.3 S-NPP-equivalent cell-days for that date from 416 MODIS cells during the documented processing gap. It is not a second sensor observation; July 2024 still has only one comparable prior year. Version-matched leave-one-year-out calibration for 2023–2026 has 41 month-pairs in each region: Northern California selects the annual ratio (median absolute log error 0.256 versus 0.923 with no scaling; interval coverage 41.5%), while Punjab–Haryana selects the monthly ratio (0.309 versus 1.723; coverage 39.0%). These intervals are substantially under-covered and remain weak. Each region has 48 complete MODIS source-months and 47 complete S-NPP source-months; worldwide S-NPP May 2026 remains unknown. Grove native masks decode 20 of 20 expected granules into 15,495 pixels and Park decodes 114 of 114 into 1,372,872 pixels; both are processed-unreviewed, with 30-cell review queues and full-footprint accounting still open. Partial historical-month verdicts expose source-specific counts and retain unknown dates. The full test suite is rerun with this refresh. The prior 1440 px headless browser check confirmed 15 sourced wildfire cases, but ArcGIS Earth stayed at its loading screen; globe-marker rendering and phone layout remain unverified.

Changelog

| 2026-09-30 | S-NPP archive completion as row evidence | Imported owner-supplied `fire_archive_SV-C2_814833.csv` with an explicit `reconstructed-rows-only` sidecar. It adds 204,694 unique clipped S-NPP rows across 26 partial month slices (July 2021–July 2022), updates the database to 1,994,162 standard rows, and exposes the previously missing period without falsely promoting it to complete coverage. Rebuilt aggregates, calibration artifacts and the static bundle; score remains **41.6 / 100** because request metadata, native independent review, MCD64A1, outside-user verification and public deployment remain open. |

When (UTC)	Who	What changed
2026-09-29	Lead engineer	Baseline audit, 31.8 / 100
2026-09-29	Owner	Globe stays. Target MODIS from 2010-07-01, Suomi NPP from 2012-07-01, across two regions; files are still downloading.
2026-09-29	Lead engineer	Plan 1.2. Withdrew claims of 54.2, 88, and 106. Remaining tasks are the required table above. Claim is 44.8 and 93, only if those checks pass.
2026-09-29	Implementation continuation	C8-T1 complete; score moves 31.8 → 33.2. README, Apache license, source ledger and AI-use disclosure updated. Two-region extension remains blocked by missing source archives; 77 tests pass.
2026-09-29	Calendar/data continuation	Region-aware import workflow and calendar are integrated. Authentic-sample held-out calibration is reproducible, raising C10 from 2 to 3; implementation score 33.2 → 34.8. This does not pass the 2012 step or full-period two-region data gates.
2026-09-29	Archive import continuation	Imported all eight matching-hash standard archive CSVs into both regions on a staging copy, corrected the global-empty S-NPP May 2026 state to unknown, validated the copy, then promoted it locally. Two-region 2022–2026 data now render; the 2010/2012 target and independent validation gates remain open, so the local score stays 34.8.
2026-09-29	Long-history display continuation	Calendar now uses the frozen July 2010 MODIS / July 2012 S-NPP start, draws pre-window dates as non-selectable blanks, and preserves NASA outage patterns even when the harmonized value is unknown. Import instructions cover contiguous year-sized requests. Existing local archive is unchanged; C1-T2 remains partial and score stays 34.8 until the older files are imported.
2026-09-29	Legacy FIRMS import continuation	Processed 27 supplied standard archive CSVs: eight with saved request metadata and 19 older files using locally generated row-only sidecars. Added 1,322,659 older rows across both regions. API checks show 2010-07 and 2012-07 detections as partial, 2022–2026 verified month counts unchanged, and the score remains 34.8 because completeness and independent-review gates are still open.
2026-09-29	Paired daily bundle continuation	Built and linked two source-hashed daily bundles for 47 complete paired months per region. All 31 July 2024 MODIS values match the existing calendar in both regions after type-2 rows were consistently excluded. C2-T2 is DONE and C2 moves 3→4; implementation score 34.8→37.2. Park masks remain unprocessed and Grove lacks one geolocation companion.
2026-09-29	Outage-ledger check	Confirmed C2-T3 acceptance: the dated NASA notice has a source URL and short quote, interval endpoints are tested, and the UI reads outage state from the ledger. Marked DONE; score remains 37.2 because C2 is already 4/5.
2026-09-29	Calibration and calendar continuation	Added per-region calibration JSON with parent and bundle hashes, and corrected annual held-out errors to use only complete 12-month years. Partial historical-month verdicts now show sensor-specific counts and unknown dates. Reprocessed supplied Grove files with the workstation GDAL stack: 19/20 granules, 15,293 pixels; one geolocation companion is missing. All 101 tests pass. Score remains 37.2: weak interval coverage, pending 2012 coverage, missing Grove companion, unreviewed samples, and the absent Park mask set leave the gates partial.
2026-09-29	Critical-date and verdict validation	Added synthetic acceptance tests for percentile verdict branches and critical-date thresholds; checked real 2023 critical dates through the API for both regions (zero missing days). C1 moves 3→4; implementation estimate 37.2→39.6. Full regression suite to be rerun before release.
2026-09-29	Static calendar packaging	Built and exercised a local no-database bundle with 34 calendars for 2010–2026, two histories, and daily source-row files. Headless Chrome loaded the dated archive, switched to Punjab–Haryana, and opened a 25 July source record. Added manifest checksum/size verification and static build instructions. Static globe API requests still return 404 and the bundle is not deployed; C12 moves 2→3 and the local estimate 39.6→40.4. The 107-test suite passes.

When (UTC)	Who	What changed
2026-09-29	Historical calendar evidence	Replaced a nearly blank C1-T2 screenshot with a 2880×2800 capture from the local live archive. It shows the full 2010–2026 history, unknown/reconstructed months, and July 2024 day tiles where 24 and 29 are visibly distinguished as partial outage days. Headless browser inspection confirmed 6,028 rendered daily history cells and accessible date/value labels. C1-T2 is DONE; the category remains 4/5 because independent scientific review is still open.
2026-09-29	Verdict-first calendar	Moved the API-backed verdict and official-source links above the calendar heading. A live browser comparison matched the Northern California July 2024 sentence to <code>/api/v2/calendar</code> ; the source links render beneath it. C9 moves 2→3, adding 1.2 points; implementation score 40.4→41.6.
2026-09-29	Authentic acceptance checks	Added regressions for all 31 July 2024 MODIS day counts, the 12-month 2024 API response, and original 25 July source rows with the documented S-NPP gap. Marked only acceptance checks with test or data evidence. The full suite passes 108 tests. Implementation remains 41.6 / 100 : the next score-bearing gates still require verified older request windows, the complete 2012 overlap, independent scientific review, or public demo readiness.
2026-09-29	Submission preparation	Prepared a verified 150-word project-page draft and aligned the video storyboard to the plan's 30-second format. Added a copy-ready, unsent Local Lead question for the pre-event reuse gate. No submission or public deployment occurred; the implementation score remains 41.6 / 100 and G0 remains blocked.
2026-09-29	Calibration display cross-check	A headless-browser check matched all three model errors, annual errors, selected method, and nominal interval coverage against each region's JSON artifact; saved captures for both regions. C10 remains partial because the complete 2012 step and independent review are unresolved, so the implementation score stays 41.6 / 100 .
2026-09-29	Plan evidence recheck	Marked C10-T1 done against its checked UI-to-JSON acceptance test; the verifier now checks that incomplete 2012 input stays visibly untested with no ratio. C10-T2 remains partial because the 2012 exports are row-only. Verified Data & method → calendar keeps the same terrain Earth renderer locally and reran all 108 tests. Public deployment and human review remain open; score stays 41.6 / 100 .
2026-09-29	Landing Earth consistency correction	A browser check caught that <code>/earth.html</code> was the separate photorealistic preview, not the intended initial globe. Restored <code>/terrain-earth.html</code> for the landing page and method → calendar handoff, and synced the static bundle. Headless Chrome reports the <code>arcgis-terrain</code> renderer with no page errors. The public demo remains unverified; score stays 41.6 / 100 .
2026-09-29	Static bundle repair	Found the previous <code>site/</code> lacked <code>fireatlas-static-data</code> , so its calendar fell back to missing API routes. Rebuilt from the local database, verified the no-database calendar and method-page return in Chrome, and promoted the bundle while preserving the old copy in <code>/tmp/fireatlas-site-before-static-meta-20260929</code> . 34 calendars, 2 history files, 72 data files; public deployment and static globe snapshot endpoints remain open. Score stays 41.6 / 100 .
2026-09-29	Correct intended Earth renderer	The previous check confirmed renderer continuity but selected the wrong visual model. Restored the original <code>/earth.html?embed=landing</code> globe for the home/calendar flow and synced the static bundle. Headless Chrome clicked Data & method → Open calendar on both local servers; the orbital Earth bridge, observation overlay, Satellite signals toggle, and Wildfires toggle all loaded. The app server retained the selected day and AOI; the static page returned to its calendar anchor but does not have the same live API context. C11-T1 is done locally, but C11 remains 3 because the public-URL gate is still open; total stays 41.6 / 100 .
2026-09-29	Globe continuity follow-up	After the globe changed during the method → calendar return, restored the original terrain embed target in the app and static copy. The served JavaScript and HTTP routes were checked. A fresh headless WebGL render stalled, so current visual continuity is not verified; C11-T1 is now marked PARTIAL and C11 remains 3. No other globe work is in scope.
2026-09-29	Local demo video	Replaced the incomplete prior cut with a deterministic 30-second captioned sequence: home capture, two regional calendar verdicts, selected S-NPP gap day, source rows, held-out uncertainty, and an honest local-only end card. Verified H.264, 1600×900, 30 fps, 30.000 seconds, silent. No public URL or submission is verified; the implementation score remains 41.6 / 100 .
2026-09-29	Local rubric impact correction	Re-audit found that the earlier 70 estimate used a July Punjab–Haryana scene that did not establish the requested crop-residue monitoring context. The local estimate returned to 65 / 118 pending a dated, sourced seasonal example. The weighted implementation estimate remained 41.6 / 100 .
2026-09-29	Seasonal impact evidence	Added a date-specific Punjab–Haryana context strip for October/November 2024, linked to the Government of India's 2024 deployment dates , and stated that hotspots do not identify crop-burning cause. Replaced the July video capture with the browser-verified October calendar: 5,174 observed cell-days and one comparable year. The new <code>scripts/verify_season_context.py</code> passes for October, November, out-of-window months, other regions, and 1440/768/390 px widths. The plan's Impact gate is now evidenced; local rubric estimate 65 → 70 / 118 . This is not a NASA judge score; the weighted implementation estimate remains 41.6 / 100 .
2026-09-30	Static evidence continuation	Exported dated Park/Grove validity reports, evidence ZIPs and analytical recount JSON into the static bundle. The method page now scrubs case days, opens source rows and recounts without an API; the Data Sources page reads the regional ledger from <code>regions.json</code> and disables server-only imports in a static copy. Park media and Grove's no-photo official record were browser-checked. C12 remains partial because no public URL is verified; the implementation estimate stays 41.6 / 100 .
2026-09-30	Native review workflow	Added a hash-bound <code>fireatlas.mask_review</code> template/validator for the 30 native samples. It requires explicit outcomes, observed classes, disagreement notes, reviewer metadata and matching input hashes; installation stays local and the UI gates remain pending until a real independent reviewer supplies the record. The score remains 41.6 / 100 .
2026-09-30	Evidence reproducibility hardening	Rounded only displayed native grid-cell centre coordinates to six decimals so ZIP reports reproduce across projection-library builds. Rebuilt <code>site/</code> with both review templates and verified Park/Grove ZIPs under system Python plus all five no-API static browser checks. Native masks and independent review remain incomplete; score remains 41.6 / 100 .
2026-09-30	Review form made judge-visible	Added a standalone download beside each case's evidence ZIP and a matching <code>/api/validity/review-template</code> response. The form is hash-bound and blank; it makes the pending human gate executable without pretending that a reviewer has signed off. Static/browser checks and the 109-test suite pass; score remains 41.6 / 100 .

When (UTC)	Who	What changed
2026-09-30	Perfected plan adopted	Compared the supplied updated PDF with the earlier plan. The authoritative path now requires common 1 km VIIRS-equivalent cell-days, native 375 m detail, visible bridge states, source-separated FRP, explicit <code>evidence_state</code> , excluded-type counts, optional lagged MCD64A1 corroboration, a share card, and a 90-second story. No numeric score uplift is assigned: MCD64A1, complete P0 windows, independent review, external usability, public deployment, and the 90-second rehearsal remain open.
2026-09-30	Perfected release refresh	Rebuilt the static bundle from the current calendar code (34 region-year calendars, 97 manifest-checked files), verified documented-gap labels, 200 bundled source rows, the share card, both validity cases, the two-region data page, and unchanged globe shell. The live API now returns separate <code>evidence_state=scaled</code> and <code>coverage_state=documented_processing_gap</code> for outage estimates. Full suite: 109 tests. Score remains 41.6 / 100 because the open P0/P1 gates are unchanged.
2026-09-30	Perfected traceability and local rehearsal	Added a visible bundle/build line, exact bridge transform label, and abbreviated SHA-256 inputs to the share card. Rebuilt the bundle with 34 region-year calendars and 98 generated data files; captured bridge, source-drawer, method, and share-card frames from that same bundle. Rendered and verified <code>FireAtlas_90s_perfected_demo.mp4</code> (1600×900, 30 fps, silent, ~90 s). Public deployment, external usability, MCD64A1, native-mask review, and eligibility ruling remain open; score stays 41.6 / 100 .
2026-09-30	Perfected visual grammar and rehearsal correction	Aligned calendar and history states with the perfected plan's color-independent grammar: observed cyan solid, estimated amber hatch, documented gap purple hatch, unknown gray crosshatch, and explicit typical/high activity keys. The monthly activity card now carries a machine-readable and accessible baseline flag. Corrected SUB-T2 to record the locally verified 90-second H.264 rehearsal; public deployment and outside-user gates remain open, so score stays 41.6 / 100 .
2026-09-30	Native-mask gate refresh	Reprocessed the supplied Grove HDF4/netCDF files with the workstation GDAL stack after the offline environment had recorded them as dependency failures. The ledger now records 19/20 processed granules, 15,293 clipped native samples, and 7/7 FIRMS rows reconciled (100%, target ≥98%); the missing MOD03 companion, 30-cell human review, Park masks, and full-footprint pass/cloud denominator remain open. Regenerated the static bundle with the updated evidence; score remains 41.6 / 100 because the independent and completeness gates are not passed.
2026-09-30	Native status consistency check	Added <code>scripts/verify_native_mask_status.py</code> and strengthened the static method verifier to fail on stale Park/Grove gate counts. The check passes for Grove 19/20 and 7/7, Park 0/114, and pending human review; this improves release reproducibility but does not change the score because the scientific gates remain open.
2026-09-30	Evidence-frame refresh	Re-captured the Grove method audit after the available-row wording change and re-rendered the 90-second local rehearsal. The manifest now binds the current frame (<code>2b5b5e...</code>) and video (<code>0b7ffa...</code>), still 1600×900, 30 fps, 90 seconds; score remains 41.6 / 100 because this is a local presentation refresh, not independent evidence or public deployment.
2026-09-30	Reviewer workflow	Added <code>/review.html</code> , a browser form for the deterministic 30-cell native-mask queue. It requires independent attestation, observed class/coordinates/grid, explicit outcomes, and preserves source hashes before downloading the JSON consumed by <code>fireatlas.mask_review</code> . Grove loads 30 samples; Park correctly reports no queue. This removes friction for the human gate but does not count as a review or raise the score until a real reviewer signs and the Park/native completeness gates are resolved.
2026-09-30	Archive and history refresh	Imported six owner-supplied MODIS archive tables with explicit <code>reconstructed-rows-only</code> provenance. Standard MODIS history now has 456,202 rows and reaches positive detections from July 2006; the verified complete 2022–2026 window is unchanged and the S-NPP 2021–2022 nominal window remains absent. Reprocessed native evidence now records Grove 20/20 and 15,495 clipped pixels plus Park 114/114 and 1,372,872 clipped pixels, with both independent review and full-footprint gates still open. Rebuilt the static bundle for the 2006–2026 display; score remains 41.6 / 100 .
2026-09-30	Share-card evidence refresh	Added JSON-derived quality/coverage states, the active-fire limitation, and an exact no-credential URL to the share card. Rebuilt the static bundle with manifest SHA-256 <code>a4b14a4e3d73ce404608d7d763cd43ebf8170ad8c67192541d791c64f966dec1</code> (42 region-year calendars, 113 manifest-listed data files, 830 input hashes). Static calendar, method, data, shell, responsive-season, native-status, compile, diff, and 111-test checks pass; public deployment, outside-user verification, MCD64A1, and human native review remain open, so score stays 41.6 / 100 .
2026-09-30	Perfected rehearsal evidence refresh	Re-captured the share card with visible quality/coverage states, ACTIVE FIRE ONLY limitation, and exact URL; refreshed Grove's method frame to the current 20/20 granules, 15,495 samples, and 7/7 available FIRMS rows matched. Re-rendered and verified the local 90-second H.264 rehearsal; its manifest binds the current frame and video hashes (1600×900, 30 fps, silent). This presentation update does not resolve the public deployment or human/scientific gates; score remains 41.6 / 100 .
2026-09-30	S-NPP-backed rehearsal refresh	Re-captured the two changing rehearsal frames from the promoted static bundle and re-rendered the 90-second H.264 cut. The current static manifest is <code>48b1065e...</code> (42 calendars, 113 manifest-listed data files, 854 input-file hashes); the rehearsal manifest binds the share-card frame <code>e53d5a84...</code> , Grove frame <code>125c5580...</code> , and video <code>a114f164...</code> . Static calendar, method, data, shell and responsive-season checks remain passing; public deployment, outside-user verification, MCD64A1, and human native review remain open, so score stays 41.6 / 100 .
2026-09-30	Static release hardening and action paths	Confirmed all 40 rows in <code>NASA_data/fire_masks/download_checklist.csv</code> have local files. Compacted browser validity snapshots while retaining complete evidence ZIPs, reducing the rebuilt static bundle from about 443 MB to 177 MB. Added the GitHub Pages workflow and <code>.nojekyll</code> , unignored the publishable <code>site/data</code> subtree, exposed three role-based action paths, and hash-bound the static bundle manifest plus a timed WebVTT caption sidecar to the 90-second local rehearsal. The current bundle has 42 region-year calendars (2006–2026), 113 checksummed data files and 854 input hashes; manifest SHA-256 is <code>e74a06f4...</code> . Local static, season, calibration, native-status, JavaScript, manifest, and full 113-test checks pass; public deployment, outside-user verification, MCD64A1, and human native review remain open, so score stays 41.6 / 100 .